United States Patent Application Publication	20250270692
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	KADOBE; Masato et al.

SUBSTRATE PROCESSING APPARATUS

Abstract

A substrate processing apparatus, includes: a reaction tube; a gas introducer configured to introduce gas to an interior of the reaction tube; a gas exhauster configured to exhaust the gas introduced to the interior of the reaction tube; a first heater configured to heat the reaction tube; a second heater configured to heat the gas exhauster; and a housing configured to accommodate the reaction tube, the gas introducer, the gas exhauster, the first heater, and the second heater in an interior of the housing.

Inventors: **KADOBE; Masato (Tokyo, JP), YAMAZAKI; Mitsuru (Iwate, JP)**

Applicant: **TOKYO ELECTRON LIMITED** (Tokyo, JP)

Family ID: **1000008502758**

Appl. No.: **19/054543**

Filed: **February 14, 2025**

Foreign Application Priority Data

JP	2024-025083	Feb. 22, 2024

Publication Classification

Int. Cl.: **C23C16/455** (20060101); **C23C16/44** (20060101); **C23C16/46** (20060101)

U.S. Cl.:

CPC **C23C16/455** (20130101); **C23C16/4412** (20130101); **C23C16/46** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-025083, filed on Feb. 22, 2024, the entire contents of which are incorporated herein by reference.
TECHNICAL FIELD
[0002] The present disclosure relates to a substrate processing apparatus.
BACKGROUND
[0003] A batch-type substrate processing apparatus that collectively processes a plurality of substrates is known (see, for example, Patent Documents 1 and 2). In the substrate processing apparatus disclosed in Patent Document 1, an exhaust pipe is installed below a reaction tube, and a heater terminal is provided in a vertical side or a side surface of a heat reflector. In the substrate processing apparatus disclosed in Patent Document 2, a gas introduction pipe and a gas exhaust pipe are provided at opposing positions on a side surface of a reaction tube.
Prior Art Documents
Patent Documents
[0004] Patent Document 1: Japanese Patent Laid-Open Publication No. 2001-210631
[0005] Patent Document 2: Japanese Patent Laid-Open Publication No. 2008-172205
SUMMARY
[0006] According to one embodiment of the present disclosure, there is provided a substrate processing apparatus, includes: a reaction tube; a gas introducer configured to introduce gas to an interior of the reaction tube; a gas exhauster configured to exhaust the gas introduced to the interior of the reaction tube; a first heater configured to heat the reaction tube; a second heater configured to heat the gas exhauster; and a housing configured to accommodate the reaction tube, the gas introducer, the gas exhauster, the first heater, and the second heater in an interior of the housing.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0007] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the present disclosure, and together with the general description given above and the detailed description of the embodiments given below, serve to explain the principles of the present disclosure.
[0008] FIG. **1** is a perspective view 1 showing a substrate processing apparatus according to an embodiment.
[0009] FIG. **2** is a perspective view 2 showing the substrate processing apparatus according to the embodiment.
[0010] FIG. **3** is a vertical cross-sectional view showing the substrate processing apparatus according to the embodiment.
[0011] FIG. **4** is a horizontal cross-sectional view 1 showing the substrate processing apparatus according to the embodiment.
[0012] FIG. **5** is a horizontal cross-sectional view 2 showing the substrate processing apparatus according to the embodiment.
[0013] FIG. **6** is a view 1 showing an arrangement of side heaters.
[0014] FIG. **7** is a view 2 showing the arrangement of the side heaters.
DETAILED DESCRIPTION
[0015] Reference will now be made in detail to various embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be apparent to one of ordinary skill in the art that the present disclosure may be

practiced without these specific details. In other instances, well-known methods, procedures, systems, and components have not been described in detail so as not to unnecessarily obscure aspects of the various embodiments.

[0016] Hereinafter, non-limitative exemplary embodiments of the present disclosure will be described with reference to the accompanying drawings. Throughout all the accompanying drawings, the same or corresponding members or components will be denoted by the same or corresponding reference numerals, and redundant descriptions thereof will be omitted.

Substrate Processing Apparatus

[0017] A substrate processing apparatus **1** according to an embodiment is described with reference to FIGS. **1** to **5**. FIG. **1** is a perspective view showing the substrate processing apparatus **1** according to the embodiment when the substrate processing apparatus **1** is viewed obliquely from above. FIG. **2** is a perspective view showing the substrate processing apparatus **1** according to the embodiment when the substrate processing apparatus **1** is viewed obliquely from below. FIG. **3** is a vertical cross-sectional view showing the substrate processing apparatus **1** according to the embodiment. FIG. **4** is a horizontal cross-sectional view showing the substrate processing apparatus **1** according to the embodiment, taken along line IV-IV in FIG. **3** in an arrow direction. FIG. **5** is a horizontal cross-sectional view showing the substrate processing apparatus **1** according to the embodiment, taken along line V-V in FIG. **3** in an arrow direction.

[0018] The substrate processing apparatus **1** is a batch-type apparatus that collectively performs various processes on a plurality of substrates. The various processes include, for example, a film formation process for forming films on the substrates by atomic layer deposition (ALD) or chemical vapor deposition (CVD). The various processes may include an etching process for removing the films formed on the substrates.

[0019] The substrate processing apparatus **1** includes a reaction tube **10**, a gas introducer **20**, a vacuum pipe **30**, an exhaust duct **40**, a housing **50**, a heating part **60**, a depressurizer **70**, a pressurizer **80**, and an apparatus housing **90**. In FIGS. **1** and **2**, the housing **50**, the heating part **60**, the depressurizer **70**, the pressurizer **80**, and the apparatus housing **90** are not illustrated. The reaction tube **10**, the gas introducer **20**, and the exhaust duct **40** are joined together by, for example, welding, and are integrally formed. The reaction tube **10**, the vacuum pipe **30**, and exhaust duct **40** are formed of, for example, quartz.

[0020] The reaction tube **10** has a cylindrical shape with an open lower end and a ceiling. An introduction opening **10***a* and an exhaust opening **10***b* are formed in an outer wall of the reaction tube **10**.

[0021] The introduction opening **10***a* penetrates the outer wall of the reaction tube **10**. The introduction opening **10***a* is formed at positions at which gas introduction ducts **211** to **218**, which will be described later, are installed along a circumferential direction of the reaction tube **10**. A plurality of introduction openings **10***a* is formed along a vertical direction from a vicinity of an upper end to a vicinity of a lower end of the reaction tube **10** at each position along the circumferential direction of the reaction tube **10**. In this case, it is easy to uniformly supply gas to a range from the upper end to the lower end of the reaction tube **10**.

[0022] The exhaust opening **10***b* penetrates the outer wall of the reaction tube **10**. The exhaust opening **10***b* is formed at a position different from the position of the introduction opening **10***a* in the circumferential direction of the reaction tube **10**. The exhaust opening **10***b* is formed at a position at which the exhaust duct **40** is installed in the circumferential direction of the reaction tube **10**. The exhaust opening **10***b* is a rectangular opening that extends in the vertical direction from the vicinity of the upper end to the vicinity of the lower end of the reaction tube **10**. In this case, it is easy to uniformly exhaust gas over a range from the upper end to the lower end of the reaction tube **10**.

[0023] A lower end opening of the reaction tube **10** is hermetically sealed by a cover (not shown). The cover is made of, for example, metal such as stainless steel. A substrate holder **11** (FIG. **3**) is

accommodated inside the reaction tube **10**. The substrate holder **11** holds a plurality of substrates in a horizontal posture arranged in multiple stages in the vertical direction. The number of substrates is not limited, but is, for example, between **25** and **200**. The substrate holder **11** is made of, for example, quartz.
[0024] The gas introducer **20** includes gas introduction ducts **211** to **218**, nozzles **221** to **228**, gas introduction pipes **231** to **238**, and opening/closing valves **241** to **248**. In FIG. **3**, the gas introduction pipes **231** to **235** and the opening/closing valves **241** to **245** are shown.
[0025] The gas introduction ducts **211** to **218** are installed along the circumferential direction of the reaction tube **10**. The gas introduction ducts **211** to **218** are installed to be spaced apart from each other in the circumferential direction of the reaction tube **10**. Thereby, the adjacent gas introduction ducts **211** to **218** may have a reduced thermal effect on each other. Therefore, it is possible to suppress a temperature drop within the gas introduction ducts **211** to **218** and suppress the generation of particles. The gas introduction ducts **211** to **218** are radially installed at intervals in the circumferential direction of the reaction tube **10**. In this case, as indicated by arrows in FIG. **5**, gases such as a raw material gas, a reaction gas, an etching gas, and a purge gas may be supplied to the inside of the reaction tube **10** from a plurality of positions (in multiple directions) in the circumferential direction of the reaction tube **10**. Thereby, an in-plane shape of a film formation or etching process may be easily adjusted. For example, a retention time of gas supplied to surfaces of the substrates or a gas concentration distribution may be adjusted by adjusting a gas supply position or the amount of gas supplied. Therefore, it is easy to control the in-plane shape of the film formation or etching process compared to a unidirectional gas flow. The gas introduction ducts **211** to **218** may be installed, for example, in this order counterclockwise from the exhaust opening **10***b*.
[0026] The gas introduction ducts **211** to **218** are installed on the outer wall of the reaction tube **10**. In this case, a distance from the gas introduction ducts **211** to **218** to the substrates is shortened. Therefore, unnecessary thermal decomposition of the gas may be suppressed. In addition, when the gas introduction ducts **211** to **218** are installed on the outer wall of the reaction tube **10**, there is no need to dispose the nozzles **221** to **228** inside the reaction tube **10**. For this reason, a space between outer edges of the substrates and an inner wall of the reaction tube **10** may be narrowed, thereby suppressing a gas flow into the space. As a result, the efficiency of gas supply between the vertically adjacent substrates is improved. Additionally, when the gas introduction ducts **211** to **218** are installed on the outer wall of the reaction tube **10**, there is no need to form a nozzle chamber for accommodating the nozzles **221** to **228** by protruding a side wall of the reaction tube **10** radially outward. The gas introduction ducts **211** to **218** are integrally formed with, for example, the reaction tube **10**. The gas introduction ducts **211** to **218** are formed of, for example, quartz.
[0027] The gas introduction ducts **211** to **218** have a tubular shape with a closed lower end and an open upper end. The upper ends of the gas introduction ducts **211** to **218** extend upward from an upper surface of the reaction tube **10** and penetrate the housing **50**. In this case, an upper space of the housing **50** may be used as a space for installing the gas introduction pipes **231** to **238** or the opening/closing valves **241** to **248**. Therefore, a pipe distance from the opening/closing valves **241** to **248** to the reaction tube **10** may be shortened. Further, the shape of the gas introduction pipes **231** to **238** may be simplified. Gas holes **211***a* to **218***a* (FIG. **5**) are formed at positions of the gas introduction ducts **211** to **218** facing the reaction tube **10**.
[0028] Each of the gas holes **211***a* to **218***a* has a rectangular shape extending in the vertical direction from the vicinity of the upper end to the vicinity of the lower end of the reaction tube **10**. For example, each of the gas holes **211***a* to **218***a* extends from above the uppermost introduction opening **10***a* to below the lowermost introduction opening **10***a*. Gas flowing inside the gas introduction ducts **211** to **218** is discharged into the reaction tube **10** from the gas holes **211***a* to **218***a*.
[0029] The gas introduction duct **211** is installed in an angle range less than 90 degrees counterclockwise from the exhaust duct **40** in the circumferential direction of the reaction tube **10**.

The gas introduction duct **212** is installed at an angle of 90 degrees counterclockwise from the exhaust duct **40** in the circumferential direction of the reaction tube **10**. The gas introduction ducts **213** and **214** are installed at an angle range greater than 90 degrees and less than 180 degrees counterclockwise from the exhaust duct **40**. The gas introduction duct **215** is installed at an angle of 180 degrees counterclockwise from the exhaust duct **40** in the circumferential direction of the reaction tube **10**. In other words, the gas introduction duct **215** is installed at a position opposite to the exhaust duct **40**. The gas introduction ducts **216** and **217** are installed at an angle range greater than 180 degrees and less than 270 degrees counterclockwise from the exhaust duct **40**. The gas introduction duct **218** is installed at an angle of 270 degrees counterclockwise from the exhaust duct **40**. In other words, the gas introduction duct **218** is installed at a position opposite to the gas introduction duct **212**.

[0030] The nozzles **221** to **228** are detachably inserted into the interior of the gas introduction ducts **211** to **218**. In this case, the shape of the nozzles **221** to **228** may be changed according to the type of process, so that the optimal nozzles **221** to **228** may be used according to the type of process. An inner surface of each of the gas introduction ducts **211** to **218** has a shape according to an outer surface of each of the nozzles **221** to **228**, and a gap is provided between the inner surface of each of the gas introduction ducts **211** to **218** and the outer surface of each of the nozzles **221** to **228**. This allows a volume of a space between the gas introduction ducts **211** to **218** and the nozzles **221** to **228** to be reduced. Therefore, the retention of gas in the space is suppressed, thereby improving the efficiency of gas supply to the substrates. In addition, since a surface area in contact with gas is reduced, the generation of particles in the space may be suppressed. Thus, the space may be easily cleaned. In a cross-section perpendicular to a longitudinal direction of the nozzles **221** to **228**, the inner surface of the gas introduction ducts **211** to **218** is, for example, circular, and the outer surface of the nozzles **221** to **228** is, for example, circular. In the cross-section perpendicular to the longitudinal direction of the nozzles **221** to **228**, the inner surface of the gas introduction ducts **211** to **218** may be elliptical, and the outer surface of the nozzles **221** to **228** may also be elliptical.

[0031] The nozzles **221** to **228** are provided with gas discharge holes (not shown). The gas discharge holes are provided, for example, in portions of tube walls of the nozzles **221** to **228** inserted into the gas introduction ducts **211** to **218**. Upper ends of the nozzles **221** to **228** are connected to gas sources (not shown) via the gas introduction pipes **231** to **238**, respectively. Gases from the gas sources are introduced into the nozzles **221** to **228** from the upper ends of the nozzles **221** to **228** and discharged into the reaction tube **10** through the gas discharge holes, the gas holes **211***a* to **218***a*, and the introduction openings **10***a*. The nozzles **221** to **228** are not illustrated in FIGS. **2** to **5**.

[0032] The nozzles **221** to **228** may not be provided. In this case, the upper ends of the gas introduction ducts **211** to **218** are connected to the gas sources via the gas introduction pipes **231** to **238**, respectively. Gases from the gas sources are introduced into the gas introduction ducts **211** to **218** from the upper ends of the gas introduction ducts **211** to **218** and discharged into the reaction tube **10** through the gas holes **211***a* to **218***a*. For example, when gases that are easily thermally decomposed, such as hexachlorodisilane (HCD) gas and dichlorosilane (DCS) gas, are used, the nozzles **221** to **228** may not be necessary.

[0033] The gas introduction pipes **231** to **238** are installed in an upper space of the housing **50**. One end of each of the gas introduction pipes **231** to **238** is connected to each of the corresponding gas introduction ducts **211** to **218** or the corresponding nozzles **221** to **228**, and the other end of each of the gas introduction pipes **231** to **238** passes through the apparatus housing **90** and extends to an exterior of the apparatus housing **90**. The gas introduction pipes **231** to **238** are provided with the opening/closing valves **241** to **248**. The gas introduction pipes **231** to **238** may also be provided with flow rate controllers such as mass flow controllers.

[0034] The opening/closing valves **241** to **248** are installed in the upper space of the housing **50**. The opening/closing valves **241** to **248** are installed in the middle of the corresponding gas introduction pipes **231** to **238**. The opening/closing valves **241** to **248** are installed, for example, on an inner wall of the apparatus housing **90**. The opening/closing valves **241** to **248** are valves that switch a gas flow on and off.

[0035] The vacuum pipe **30** has a cylindrical shape with an open lower end and a ceiling. The vacuum pipe **30** is installed spaced apart from the reaction tube **10**. An opening **30***a* (FIG. **5**) is formed on an outer wall of the vacuum pipe **30** at a same position as the exhaust duct **40** in a circumferential direction of the vacuum pipe **30**. The opening **30***a* is a rectangular opening extending in the vertical direction from a vicinity of an upper end to a vicinity of a lower end of the vacuum pipe **30**. A vertical length of the opening **30***a* may be the same as a vertical length of the exhaust opening **10***b*. An axis of the vacuum pipe **30** may be parallel to an axis of the reaction tube **10**. The lower end of the vacuum pipe **30** is connected to an exhaust device (not shown) such as a vacuum pump via a pipe (not shown). A cross-sectional area of a flow path of the vacuum pipe **30** may be equal to or greater than a cross-sectional area of a flow path of the exhaust duct **40**. In this case, exhaust flow velocity in the vertical direction becomes uniform, so that a uniform laminar flow in the vertical direction is formed. Thereby, inter-plane uniformity of the film formation or etching process is improved.

[0036] The exhaust duct **40** connects the reaction tube **10** and the vacuum pipe **30**. The exhaust duct **40** allows communication between an interior of the reaction tube **10** and an interior of the vacuum pipe **30**. One end of the exhaust duct **40** is connected to the outer wall of the reaction tube **10** to cover the exhaust opening **10***b*, and the other end of the exhaust duct **40** is connected to the outer wall of the vacuum pipe **30** to cover the opening **30***a*. In this case, compared to directly connecting the vacuum pipe **30** to the reaction tube **10** without providing the exhaust duct **40**, a length X (FIG. **4**) occupied by the exhaust duct **40** in the circumferential direction of the reaction tube **10** may be shortened. As a result, a length in which the gas introduction ducts **211** to **218** may be installed in the circumferential direction of the reaction tube **10** is lengthened. Thereby, the number of the gas introduction ducts **211** to **218** installed on the outer wall of the reaction tube **10** may be increased. The exhaust duct **40** may be divided into a plurality of parts in the vertical direction. In this case, a gas flow from the interior of the reaction tube **10** towards the vacuum pipe **30** is rectified. For this reason, the uniformity of the gas flow at different positions in the vertical direction inside the reaction tube **10** is improved.

[0037] The housing **50** accommodates the reaction tube **10**, the gas introducer **20**, the vacuum pipe **30**, the exhaust duct **40**, and the heating part **60** therein. Since the housing **50** accommodates the heating part **60** including a heater, the housing **50** is also called a heater shell. The housing **50** includes a bottom portion **51**, a ceiling portion **52**, and a side portion **53**.

[0038] The bottom portion **51** supports the reaction tube **10** and the vacuum pipe **30**. The bottom portion **51** is made of, for example, stainless steel. In this case, strength for supporting the reaction tube **10** and the vacuum pipe **30** is easily secured. The bottom portion **51** is, for example, a mirror-polished surface with a mirror-polished inner surface **51***f* (FIG. **6**). In this case, it is easy to reflect thermal radiation.

[0039] The ceiling portion **52** is provided above the upper surface of the reaction tube **10** and an upper surface of the vacuum pipe **30**. The ceiling portion **52** covers the upper surface of the reaction tube **10** and the upper surface of the vacuum pipe **30**. The ceiling portion **52** is formed of, for example, stainless steel. In this case, high strength is obtained. The ceiling portion **52** is, for example, a mirror-polished surface with a mirror-polished inner surface **52***f* (FIG. **6**). In this case, it is easy to reflect thermal radiation. The ceiling portion **52** may be formed of aluminum with the inner surface **52***f* which is a machined surface. Since the machined surface of aluminum easily reflects thermal radiation, a high reflectivity for thermal radiation is obtained without mirror-polishing the inner surface **52***f*. Therefore, the housing **50** may be manufactured at low cost. The ceiling portion **52** may be provided with a refrigerant flow path **52***g* (FIG. **6**) for circulating a refrigerant. In this case, the ceiling portion **52** may be cooled by circulating the refrigerant through the refrigerant flow path **52***g*. In addition, since the upper surface of the reaction tube **10** is covered with the inner surface **52***f* of the cooled ceiling portion **52**, temperature responsiveness during cooling of the reaction tube **10** may be enhanced even when the interior of the housing **50** is in a depressurized state.

[0040] The side portion **53** is provided around the reaction tube **10**, the gas introducer **20**, the vacuum pipe **30**, and the exhaust duct **40**. The side portion **53** covers the reaction tube **10**, the gas introducer **20**, the vacuum pipe **30**, and the exhaust duct **40**. A lower end of the side portion **53** is connected to the bottom portion **51**, and an upper end of the side portion **53** is connected to the ceiling portion **52**. The side portion **53** may be formed of, for example, aluminum with an inner surface **53***f* (FIG. **6**) which is a machined surface. Since the machined surface of aluminum easily reflects thermal radiation, a high reflectivity for thermal radiation may be obtained without mirror-polishing the inner surface **53***f*. Therefore, the housing **50** may be manufactured at low cost. The side portion **53** may be provided with a refrigerant flow path **53***g* (FIG. **6**) for circulating a refrigerant. In this case, the side portion **53** may be cooled by circulating the refrigerant through the refrigerant flow path **53***g*. In addition, since the side surface of the reaction tube **10** is covered with the cooled inner surface **53***f*, temperature responsiveness during cooling of the reaction tube **10** may be enhanced even when the interior of the housing **50** is in a depressurized state.

[0041] In this way, when the inner surfaces **51***f*, **52***f*, and **53***f* are the mirror-polished surfaces or machined surfaces, the thermal radiation from the heating part **60** may be reflected from the entire surfaces (the inner surfaces **51***f*, **52***f*, and **53***f*) of the housing **50** to uniformly heat the substrates.

[0042] The bottom portion **51**, the ceiling portion **52**, and the side portion **53** may be configured as separate bodies. The bottom portion **51**, the ceiling portion **52**, and the side portion **53** may be configured as a single body.

[0043] The heating part **60** is installed inside the housing **50**. The heating part **60** includes a first side heater **61**, a second side heater **62**, a third side heater **63**, a first ceiling heater **64**, a second ceiling heater **65**, and a lower heater **66**. The first side heater **61**, the second side heater **62**, the third side heater **63**, the first ceiling heaters **64**, the second ceiling heaters **65**, and the lower heater **66** are, for example, carbon wire heaters. In this case, the substrates accommodated inside the reaction tube **10** may be rapidly heated and cooled.

[0044] A plurality of first side heaters **61** is installed around the reaction tube **10**. The plurality of first side heaters **61** is radially installed at intervals in the circumferential direction of the reaction tube **10**. Each of the first side heaters **61** is positioned at a different location from the exhaust duct **40** in the circumferential direction of the reaction tube **10**. Each of the first side heaters **61** may be divided into a plurality of parts in the vertical direction. In this case, temperature in the vertical direction may be independently adjusted by independently controlling the first side heaters **61** divided into the plurality of parts. As indicated by solid arrows in FIG. **4**, the first side heaters **61** heat the substrates accommodated inside the reaction tube **10** from the exterior of the reaction tube **10** using thermal radiation.

[0045] The second side heater **62** is installed in a different position from the position of the first side heater **61** in the circumferential direction of the reaction tube **10**. The second side heater **62** is installed in a different position from the positions of the plurality of gas introduction ducts **211** to **218** in the circumferential direction of the reaction tube **10**. The second side heater **62** is installed in a position including a same position as the exhaust duct **40** in the circumferential direction of the reaction tube **10**. Since the exhaust duct **40** is installed in a periphery of the reaction tube **10** at the same position as the exhaust duct **40** in the circumferential direction of the reaction tube **10**, it is not possible to locate the second side heater **62**. Therefore, the second side heater **62** is installed around the vacuum pipe **30**. In other words, the second side heater **62** is installed at a position to which a distance from a center C**1** of the reaction tube **10** is farther than a distance from the center C**1** to the first side heater **61**. For example, the second side heater **62**, when viewed in a plane, is installed to include a virtual half straight line L extending through a center C**3** of the vacuum pipe

30 starting from the center C**1** of the reaction tube **10**. The second side heater **62** may be divided into a plurality of parts in the vertical direction. In this case, temperature in the vertical direction may be independently adjusted by independently controlling the second side heater **62** divided into the plurality of parts. As indicated by a broken line arrow in FIG. **4**, the second side heater **62** heats the vacuum pipe **30** and the exhaust duct **40** using thermal radiation and also heats the substrates accommodated inside the reaction tube **10**. Thereby, the substrates accommodated inside the reaction tube **10** is heated from all directions around the reaction tube **10** by the first side heater **61** and the second side heater **62**. Therefore, the in-plane temperature uniformity of a substrate is improved.
[0046] A plurality of third side heaters **63** is installed around the vacuum pipe **30**. The plurality of third side heaters **63** is installed at intervals in the circumferential direction of the vacuum pipe **30**. Each of the third side heaters **63** is positioned at a position different from the position of the second side heater **62** in the circumferential direction of the vacuum pipe **30**. Each of the third side heaters **63**, when viewed in a plane, is installed not to include the virtual half straight line L. Each third side heater **63** may be divided into a plurality of parts in the vertical direction. In this case, temperature in the vertical direction may be independently adjusted by independently controlling the third side heater **63** divided into the plurality of parts. As indicated by dash-dotted line arrows in FIG. **4**, the third side heaters **63** heat the vacuum pipe **30**.
[0047] The first ceiling heater **64** is installed between the upper surface of the reaction tube **10** and the ceiling portion **52** of the housing **50**. The first ceiling heater **64** heats the substrates accommodated inside the reaction tube **10** from above the reaction tube **10** using thermal radiation. The number of first ceiling heaters **64** may be one or may be two or more.
[0048] The second ceiling heater **65** is installed between the upper surface of the vacuum pipe **30** and the ceiling portion **52** of the housing **50**. The second ceiling heater **65** heats the vacuum pipe **30** from above the vacuum pipe **30** using thermal radiation. The number of second ceiling heaters **65** may be one or may be two or more.
[0049] A plurality of lower heaters **66** is installed in a periphery of a lower portion of the reaction tube **10**. The plurality of the lower heaters **66** is installed radially at intervals in the circumferential direction of the reaction tube **10**. The lower heaters **66** are positioned below the substrate holder **11**. The lower heaters **66** heat the lower portion of the reaction tube **10** using thermal radiation and suppress heat dissipation from the lower end opening of the reaction tube **10**.
[0050] The depressurizer **70** reduces pressure inside the housing **50**. The depressurizer **70** includes a pipe **71**, a safety valve **72**, an opening/closing valve **73**, and a vacuum pump **74**.
[0051] The pipe **71** is connected to a port **53***a* installed in the side portion **53** of the housing **50**. The pipe **71** penetrates through the side portion **53** and the apparatus housing **90** and extends to the exterior of the apparatus housing **90**. In the pipe **71**, the safety valve **72**, the opening/closing valve **73**, and the vacuum pump **74** are installed in this order from the housing **50**. The safety valve **72**, the opening/closing valve **73**, and the vacuum pump **74** are installed, for example, outside the apparatus housing **90**. The safety valve **72**, the opening/closing valve **73**, and the vacuum pump **74** may be installed outside the housing **50** and inside the apparatus housing **90**.
[0052] When a pressure inside the housing **50** exceeds a set pressure, a state of the safety valve **72** is changed from a closed state to an open state, thereby maintaining the internal pressure of the housing **50** below the set pressure.
[0053] The opening/closing valve **73** is a valve that switches a gas flow on and off.
[0054] The vacuum pump **74** reduces the pressure inside the housing **50** through the pipe **71**.
[0055] When the opening/closing valve **73** becomes an open state, the interior of the housing **50** is depressurized by the vacuum pump **74**. In a state in which the interior of the housing **50** is depressurized, heat transfer by convection is suppressed. This suppresses heat transfer to the exterior of the housing **50**, so that the space above the housing **50** may be used to install the opening/closing valves **241** to **248**. In addition, since a thermal insulation material becomes

unnecessary, a distance between the reaction tube **10** and the housing **50** may be shortened. Thereby, a distance from the opening/closing valves **241** to **248** to the substrates may be shortened and thus gas controllability is improved. Furthermore, when the side heaters (the first side heater **61**, the second side heater **62**, and the third side heater **63** are divided into a plurality of parts in the vertical direction, there is no influence of convection inside the housing **50**, so that heaters have little influence on one another in the vertical direction. Therefore, inter-plane (vertical direction) temperature controllability is improved. This makes it easy to selectively heat only the lower portion of the reaction tube **10**, selectively heat only the center of the reaction tube **10**, or selectively heat only the upper portion of the reaction tube **10**.

[0056] The pressurizer **80** restores the internal pressure of the depressurized housing **50** to atmospheric pressure. The pressurizer **80** includes a pipe **81**, a gas source **82**, a flow rate controller **83**, and an opening/closing valve **84**.

[0057] The pipe **81** is connected to a port **53***b* installed in the side portion **53** of the housing **50**. In the pipe **81**, the gas source **82**, the flow rate controller **83**, and the opening/closing valve **84** are installed in this order from an upstream to a downstream of a gas flow direction. The gas source **82**, the flow rate controller **83**, and the opening/closing valve **84** are installed, for example, outside the apparatus housing **90**. The gas source **82**, the flow rate controller **83**, and the opening/closing valve **84** may be installed outside the housing **50** and inside the apparatus housing **90**.

[0058] The gas source **82** is, for example, a supply source of an inert gas. The inert gas is, for example, nitrogen gas. The inert gas may be argon gas.

[0059] The flow rate controller **83** controls the flow rate of gas flowing through the pipe **81**. The flow rate controller **83** is, for example, a mass flow controller.

[0060] The opening/closing valve **84** is a valve that switches a gas flow on and off.

[0061] When the opening/closing valve **84** becomes an open state, the flow rate of gas from the gas source **82** is controlled by the flow rate controller **83**, and the gas with the controlled flow rate is supplied to the interior of the housing **50**. Thereby, the internal pressure of the depressurized housing **50** is restored to atmospheric pressure.

[0062] The apparatus housing **90** surrounds the housing **50**. The apparatus housing **90** covers

[0063] the entirety of the housing **50**. The apparatus housing **90** supports the bottom portion **51** of the housing **50**.

[0064] As described above, according to the substrate processing apparatus **1** of the embodiment, the reaction tube **10**, the gas introducer **20**, the vacuum pipe **30**, and the exhaust duct **40** are accommodated inside the housing **50**, and the heating part **60** is installed inside the housing **50**. This allows the substrates accommodated inside the reaction tube **10** to be heated from all directions around the reaction tube **10**. Thereby, the in-plane temperature uniformity of a substrate may be improved. As a result, processing uniformity of the substrate is improved.

Side Heaters

[0065] An example of side heaters (the first side heater **61**, the second side heater **62**, and the third side heater **63**) will be described with reference to FIGS. **6** and **7**. FIGS. **6** and **7** are views illustrating the arrangement of the side heaters. FIG. **6** is a vertical cross-sectional view, and FIG. **7** is a horizontal cross-sectional view. FIG. **7** shows a cross-section taken along line IV-IV in FIG. **3** in an arrow direction.

[0066] The first side heater **61** has an upper heater **611**, a central heater **612**, and a lower heater **613**. The upper heater **611**, the central heater **612**, and the lower heater **613** are, for example, carbon wire heaters.

[0067] The upper heater **611** has a heater element **611***a*, a terminal **611***b*, and a sealing portion **611***c*. The heater element **611***a* is arranged at an upper portion within the housing **50**. The heater element **611***a* has a configuration in which a high-purity carbon wire heating element is embedded within a quartz glass tube. The heater element **611***a* is formed, for example, in a wave-like shape in which a U-shape is continuously formed. The terminal **611***b* is connected to the heater element **611***a* and is drawn out above the housing **50** through the ceiling portion **52** in a sealed state by the sealing portion **611***c*. The upper heater **611** is supplied with power through the terminal **611***b*, so that the carbon wire heating element of the heater element **611***a* generates heat and heats the substrates arranged at the upper portion within the reaction tube **10**.

[0068] The central heater **612** has a heater element **612***a*, a terminal **612***b*, and a sealing portion **612***c*. The heater element **612***a* is arranged at a central portion within the housing **50**. The heater element **612***a* has a configuration in which a high-purity carbon wire heating element is embedded within a quartz glass tube. The heater element **612***a* is formed in, for example, a wave-like shape in which a U-shape is continuously formed. The terminal **612***b* is connected to the heater element **612***a* and is drawn out above the housing **50** through the ceiling portion **52** in a sealed state by the sealing portion **612***c*. The central heater **612** is supplied with power through the terminal **612***b*, so that the carbon wire heating element of the heater element **612***a* generates heat and heats the substrates arranged at the central portion within the reaction tube **10**.

[0069] The lower heater **613** has a heater element **613***a*, a terminal **613***b*, and a sealing portion **613***c*. The heater element **613***a* is arranged at a lower portion within the housing **50**. The heater element **613***a* has a configuration in which a high-purity carbon wire heating element is embedded within a quartz glass tube. The heater element **613***a* is formed, for example, in a wave-like shape in which a U shape is continuously formed. The terminal **613***b* is connected to the heater element **613***a* and is drawn out above the housing **50** through the ceiling portion **52** in a sealed state by the sealing portion **613***c*. The lower heater **613** is supplied with power through the terminal **613***b*, so that the carbon wire heating element of the heater element **613***a* generates heat and heats the substrates arranged at the lower portion within the reaction tube **10**.

[0070] The terminal **611***b* of the upper heater **611**, the terminal **612***b* of the central heater **612**, and the terminal **613***b* of the lower heater **613** may be installed at positions offset from one another in a circumferential direction of the reaction tube **10**, as shown in FIG. **7**. In this case, while ensuring the maintainability of the terminals **611***b*, **612***b*, and **613***b*, the upper heater **611**, the central heater **612**, and the lower heater **613** may be brought close to the reaction tube **10**. Therefore, the temperature controllability of the substrates accommodated inside the reaction tube **10** is improved. In addition, since the shapes of the heater elements **611***a*, **612***a*, and **613***a* may be made uniform, the manufacturing cost of the heater elements **611***a*, **612***a*, and **613***a* may be reduced.

[0071] The second side heater **62** has a heater element **62***a*, a terminal **62***b*, and a sealing portion **62***c*. The heater element **62***a* is arranged from the upper portion to the lower portion within the housing **50**. The heater element **62***a* has a configuration in which a high-purity carbon wire heating element is embedded within a quartz glass tube. The heater element **62***a* is formed, for example, in a wave-like shape in which a U-shape is continuously formed. The terminal **62***b* is connected to the heater element **62***a* and is drawn out above the housing **50** through the ceiling portion **52** in a sealed state by the sealing portion **62***c*.

[0072] The third side heater **63** has a heater element **63***a*, a terminal **63***b*, and a sealing portion **63***c*. The heater element **63***a* is arranged from the upper portion to the lower portion within the housing **50**. The heater element **63***a* has a configuration in which a high-purity carbon wire heating element is embedded within a quartz glass tube. The heater element **63***a* is formed, for example, in a wave-like shape in which a U-shape is continuously formed. The terminal **63***b* is connected to the heater element **63***a* and is drawn out above the housing **50** through the ceiling portion **52** in a sealed state by the sealing portion **63***c*.

[0073] As described above, all of the terminals **611***b*, **612***b*, **613***b*, **62***b*, and **63***b* of the side heaters (the upper heater **611**, the central heater **612**, the lower heater **613**, the second side heater **62**, and the third side heater **63**) are drawn out above the housing **50** through the ceiling portion **52**. As a result, maintenance work for all of the side heaters may be performed above the housing **50**, making maintenance work easy.

[0074] For the first ceiling heater **64**, the second ceiling heater **65**, and the lower heater **66**,

terminals are drawn out above the housing through the ceiling portion **52** in the same manner as the terminals of the side heaters. As a result, maintenance work for the first ceiling heater **64**, the second ceiling heater **65**, and the lower heater **66** may be performed above the housing **50**, making maintenance work easy. In addition, since the terminals of the heaters are not drawn out around the housing **50**, the apparatus becomes compact.
[0075] In addition, in the above embodiment, the vacuum pipe **30** and the exhaust duct **40** are examples of a gas exhauster. The first side heater **61**, the first ceiling heater **64**, and the lower heater **66** are examples of a first heater. The second side heater **62**, the third side heater **63**, and the second ceiling heater **65** are examples of a second heater. The first side heater **61** is an example of a reaction tube side heater. The first ceiling heater **64** is an example of a reaction tube ceiling heater. The second side heater **62** and the third side heater **63** are examples of an exhaust side heater. The second ceiling heater **65** is an example of an exhaust ceiling heater. The upper heater **611**, the center heater **612**, and the lower heater **613** are examples of a plurality of split heaters.
[0076] It should be noted that the embodiments disclosed herein are exemplary in all aspects and are not restrictive. The above-described embodiments may be omitted, replaced, or modified in various forms without departing from the scope and spirit of the appended claims.
[0077] According to the present disclosure in some embodiments, it is possible to raise the in-plane temperature uniformity of a substrate.
[0078] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

Claims

1. A substrate processing apparatus, comprising: a reaction tube; a gas introducer configured to introduce gas to an interior of the reaction tube; a gas exhauster configured to exhaust the gas introduced to the interior of the reaction tube; a first heater configured to heat the reaction tube; a second heater configured to heat the gas exhauster; and a housing configured to accommodate the reaction tube, the gas introducer, the gas exhauster, the first heater, and the second heater in an interior of the housing.
2. The substrate processing apparatus of claim 1, wherein the gas exhauster includes an exhaust duct formed integrally with the reaction tube and a vacuum pipe connected to the reaction tube via the exhaust duct, and wherein an axis of the vacuum pipe is parallel to an axis of the reaction tube.
3. The substrate processing apparatus of claim 2, wherein the housing includes a side portion installed around the reaction tube and the vacuum pipe, wherein the side portion is formed of aluminum, and wherein an inner surface of the side portion is a machined surface.
4. The substrate processing apparatus of claim 3, wherein a refrigerant flow path is formed in the side portion.
5. The substrate processing apparatus of claim 4, wherein the housing further includes a ceiling portion configured to cover an upper surface of the reaction tube and an upper surface of the vacuum pipe, wherein the first heater comprises: a first heater element installed inside the housing; and a first terminal connected to the first heater element and drawn out above the housing through the ceiling portion, and wherein the second heater comprises: a second heater element installed inside the housing; and a second terminal connected to the second heater element and drawn out above the housing through the ceiling portion.
6. The substrate processing apparatus of claim 4, wherein the first heater and the second heater are

carbon wire heaters.

7. The substrate processing apparatus of claim 4, wherein the first heater comprises: a reaction tube side heater configured to heat the reaction tube from around the reaction tube; and a reaction tube ceiling heater configured to heat the reaction tube from above the reaction tube, and wherein the second heater comprises: an exhaust side heater configured to heat the vacuum pipe from around the vacuum pipe; and an exhaust ceiling heater configured to heat the vacuum pipe from above the vacuum pipe.

8. The substrate processing apparatus of claim 7, wherein the reaction tube side heater includes a plurality of split heaters divided in a vertical direction, and wherein terminals of the plurality of split heaters are installed at positions offset from one another in a circumferential direction of the reaction tube.

9. The substrate processing apparatus of claim 2, wherein the housing includes a ceiling portion configured to cover an upper surface of the reaction tube and an upper surface of the vacuum pipe, wherein the first heater comprises: a first heater element installed inside the housing; and a first terminal connected to the first heater element and drawn out above the housing through the ceiling portion, and wherein the second heater comprises: a second heater element installed inside the housing; and a second terminal connected to the second heater element and drawn out above the housing through the ceiling portion.

10. The substrate processing apparatus of claim 2, wherein the first heater and the second heater are carbon wire heaters.

11. The substrate processing apparatus of claim 2, wherein the first heater comprises: a reaction tube side heater configured to heat the reaction tube from around the reaction tube; and a reaction tube ceiling heater configured to heat the reaction tube from above the reaction tube, and wherein the second heater includes: an exhaust side heater configured to heat the vacuum pipe from around the vacuum pipe; and an exhaust ceiling heater configured to heat the vacuum pipe from above the vacuum pipe.

12. The substrate processing apparatus of claim 1, further comprising a depressurizer configured to depressurize the interior of the housing.